

Academic Project Report on

Design and implement a website/web application for a store, which sells only.

Academic project submitted in fulfilment of the requirement of the degree of Bachelor of Computer Application



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CakeShop (branded as Sweet Crumbs Bakery) is a Java-based web application for an artisan bakery and online cake ordering system. Customers can browse the cake catalogue, register an account, place orders for delivery, and track order status. Administrators can manage the cake menu, view all customer orders, and update order fulfillment status.

The application uses a classic Java EE architecture:

- JSP pages for the user interface
- Java Servlets for business logic
- MySQL database for data storage
- HttpSession for login and role management

Project Objectives:

- ❖ Provide an online storefront for bakery products.
- ❖ Allow registered customers to place cake orders with delivery details.
- ❖ Restrict ordering to authenticated users only (no guest checkout).
- ❖ Enable administrators to manage cakes and customer orders.
- ❖ Track order lifecycle from Received to Delivered or Cancelled.
- ❖ Maintain customer accounts with profile and order history.


```

package controller;

import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;
import model.User;

@WebServlet("/login")
public class LoginServlet extends HttpServlet {

    private static final String ADMIN_USERNAME = "admin";
    private static final String ADMIN_PASSWORD = "admin123";

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        if ("logout".equals(request.getParameter("action"))) {
            HttpSession session = request.getSession(false);
            if (session != null) {
                session.invalidate();
            }
            response.sendRedirect(request.getContextPath() + "/login?success=Signed+out+successfully");
            return;
        }

        if (AuthUtil.isAdminLoggedIn(request)) {
            response.sendRedirect(request.getContextPath() + "/view-orders");
            return;
        }

        if (AuthUtil.isUserLoggedIn(request)) {
            response.sendRedirect(resolveRedirect(request));
            return;
        }

        request.getRequestDispatcher("login.jsp").forward(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        request.setCharacterEncoding("UTF-8");
        String role = request.getParameter("role");
        String username = request.getParameter("username");
        String password = request.getParameter("password");

        if ("admin".equals(role)) {
            if (ADMIN_USERNAME.equals(username) && ADMIN_PASSWORD.equals(password)) {
                HttpSession session = request.getSession(true);
                session.removeAttribute(AuthUtil.SESSION_USER);
                session.setAttribute(AuthUtil.SESSION_ADMIN, username);
                response.sendRedirect(request.getContextPath() + "/view-orders");
                return;
            }
            response.sendRedirect(request.getContextPath() + "/login?error=Invalid+admin+username+or+password");
        } else {
            try {
                User user = UserDao.findByUsername(username);
                if (user != null && user.getPassword().equals(password)) {
                    HttpSession session = request.getSession(true);
                    session.removeAttribute(AuthUtil.SESSION_ADMIN);
                    session.setAttribute(AuthUtil.SESSION_USER, user);
                    response.sendRedirect(resolveRedirect(request));
                    return;
                }
            } catch (Exception e) {
                e.printStackTrace();
            }
            response.sendRedirect(request.getContextPath() + "/login?error=Database+error.+Run+sql/users_table.sql+first");
            return;
        }

        response.sendRedirect(request.getContextPath() + "/login?error=Invalid+username+or+password"
            + appendRedirectParam(request));
    }

    private String appendRedirectParam(HttpServletRequest request) {
        String redirect = request.getParameter("redirect");
        if (redirect == null || redirect.isEmpty()) {
            return "";
        }
        try {
            return "&redirect=" + java.net.URLEncoder.encode(redirect, "UTF-8");
        } catch (java.io.UnsupportedEncodingException e) {
            return "";
        }
    }

    private String resolveRedirect(HttpServletRequest request) {
        String redirect = request.getParameter("redirect");
        if (redirect == null || redirect.isEmpty()) {
            return request.getContextPath() + "/";
        }
        if (redirect.contains("//") || redirect.startsWith("//")) {
            return request.getContextPath() + "/";
        }
        if (redirect.startsWith("/")) {
            return request.getContextPath() + redirect;
        }
        return request.getContextPath() + "/" + redirect;
    }
}

```

```

package controller;

import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.ArrayList;
import java.util.Arrays;
import java.util.HashMap;
import java.util.HashSet;
import java.util.List;
import java.util.Map;
import java.util.Set;
import java.util.Set;

public final class OrderDao {

    public static final String DEFAULT_STATUS = "Received";

    public static final List<String> VALID_STATUSES = Arrays.asList(
        "Received",
        "Baking",
        "Out for Delivery",
        "Delivered",
        "Cancelled"
    );

    private static final Set<String> VALID_STATUS_SET = new HashSet<>(VALID_STATUSES);

    private OrderDao() {
    }

    public static boolean isValidStatus(String status) {
        return status != null && VALID_STATUS_SET.contains(status.trim());
    }

    public static String normalizeStatus(String status) {
        if (status == null || status.trim().isEmpty()) {
            return DEFAULT_STATUS;
        }
        String trimmed = status.trim();
        return VALID_STATUS_SET.contains(trimmed) ? trimmed : DEFAULT_STATUS;
    }

    public static String statusCssClass(String status) {
        switch (normalizeStatus(status)) {
            case "Baking":
                return "status-baking";
            case "Out for Delivery":
                return "status-delivery";
            case "Delivered":
                return "status-delivered";
            case "Cancelled":
                return "status-cancelled";
            default:
                return "status-received";
        }
    }

    public static List<Map<String, Object>> getOrdersForUser(int userId) throws Exception {
        List<Map<String, Object>> orders = new ArrayList<>();
        String sql = "SELECT * FROM orders WHERE user_id = ? ORDER BY order_id DESC";

        try (Connection conn = DbUtil.getConnection();
            PreparedStatement ps = conn.prepareStatement(sql)) {
            ps.setInt(1, userId);
            try (ResultSet rs = ps.executeQuery()) {
                while (rs.next()) {
                    orders.add(mapOrderRow(rs));
                }
            }
        }
        return orders;
    }

    public static boolean updateOrderStatus(int orderId, String status) throws Exception {
        String sql = "UPDATE orders SET order_status = ? WHERE order_id = ?";
        try (Connection conn = DbUtil.getConnection();
            PreparedStatement ps = conn.prepareStatement(sql)) {
            ps.setString(1, normalizeStatus(status));
            ps.setInt(2, orderId);
            return ps.executeUpdate() == 1;
        }
    }

    public static Map<String, Object> mapOrderRow(ResultSet rs) throws Exception {
        Map<String, Object> row = new HashMap<>();
        row.put("id", rs.getInt("order_id"));
        row.put("cakeName", rs.getString("cake_name"));
        row.put("cakePrice", rs.getDouble("cake_price"));
        row.put("deliveryDate", rs.getString("delivery_date"));
        row.put("customerAddress", rs.getString("customer_address"));
        row.put("status", readStatus(rs));

        try {
            row.put("customerName", rs.getString("customer_name"));
            row.put("customerPhone", rs.getString("customer_phone"));
        } catch (Exception ignored) {
        }

        return row;
    }

    private static String readStatus(ResultSet rs) {
        try {
            String status = rs.getString("order_status");
            return normalizeStatus(status);
        } catch (Exception e) {
            return DEFAULT_STATUS;
        }
    }
}

```

```
package controller;

import java.sql.Connection;
import java.sql.DriverManager;

public final class DbUtil {

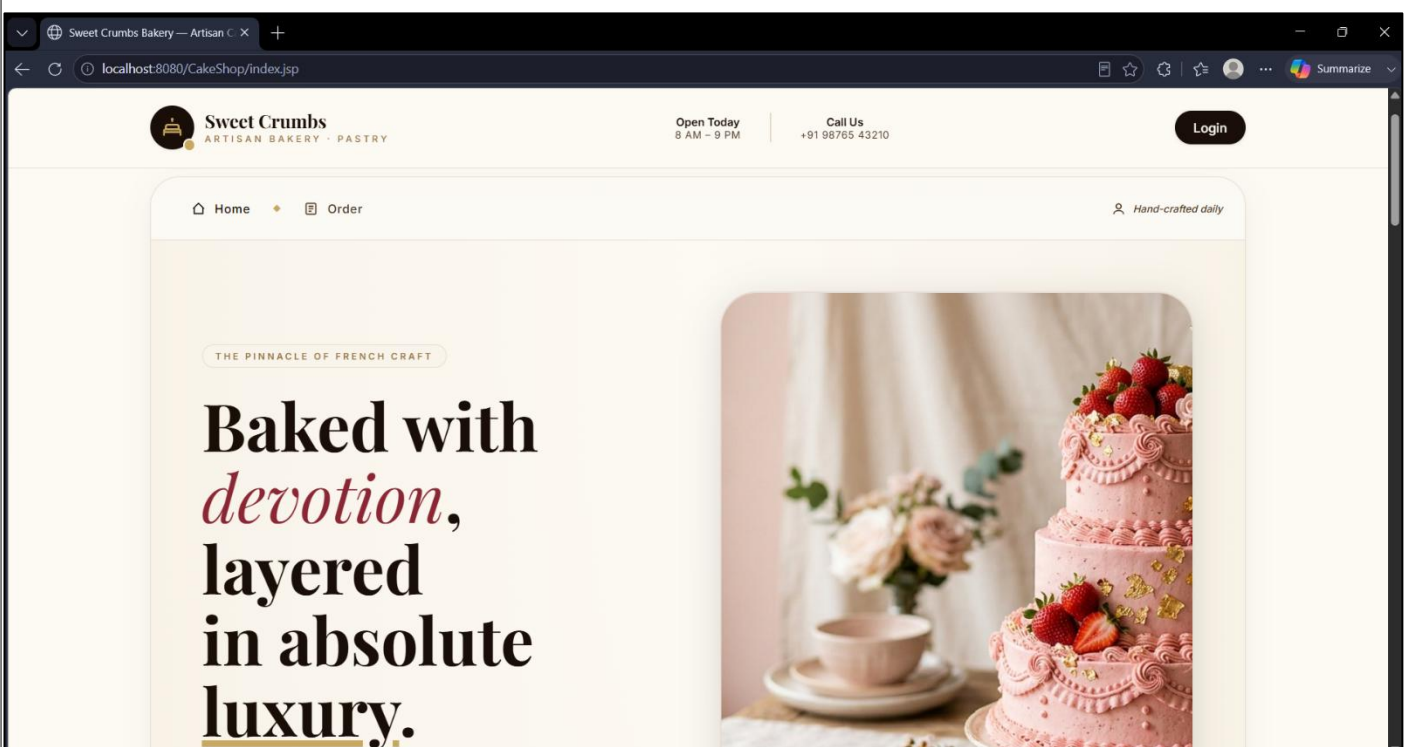
    private static final String URL =
        "jdbc:mysql://localhost:3306/sweetcrumbs_db?useSSL=false&allowPublicKeyRetrieval=true";
    private static final String USER = "root";
    private static final String PASSWORD = "102005";

    static {
        try {
            Class.forName("com.mysql.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            System.err.println("MySQL JDBC Driver not found.");
        }
    }

    private DbUtil() {
    }

    public static Connection getConnection() throws Exception {
        return DriverManager.getConnection(URL, USER, PASSWORD);
    }
}
```

Interfaces:





Sweet Crumbs
ARTISAN BAKERY · PASTRY

Open Today
8 AM - 9 PM

Call Us
+91 98765 43210

Login

The Signature *Menu*

Executed precisely to definitive, world-class perfection.



Belgian Chocolate Cake

★ 4.8

A rich and moist chocolate truffle cake layered with smooth chocolate ganache and finished with premium chocolate shavings. Perfect for birthdays, anniversaries, and special celebrations.

KEY INGREDIENTS

Chocolate Sponge Dark Chocolate Ganache
Fresh Cream Cocoa Powder
Chocolate Shavings

1 LB BASE

₹500

Buy



Red Velvet Delight

★ 4.5

A rich and moist red velvet cake layered with smooth cream cheese frosting. Its velvety texture, subtle cocoa flavor, and elegant appearance make it perfect for birthdays, anniversaries, and special celebrations.

KEY INGREDIENTS

Flour Cocoa Powder Buttermilk
Cream Cheese Butter Sugar
Eggs Vanilla Extract
Red Food Coloring

1 LB BASE

₹899

Buy



Classic Vanilla Delight

★ 4.0

A soft and fluffy vanilla sponge cake layered with rich vanilla cream and finished with a smooth, elegant frosting. Perfect for birthdays, anniversaries, and everyday celebrations. Its classic flavor and light texture make it a favorite for all age groups.

KEY INGREDIENTS

Vanilla Extract Flour Sugar
Fresh Cream Butter Eggs Milk
Baking Powder

1 LB BASE

₹399

Buy



Mango Delight

★ 4.2



Chocolate Truffle

★ 4.9

SWEET CRUMBS CLUB

Sign in & order

Create an account to track orders, save your details at checkout, and enjoy a smoother bakery experience.

[Sign In](#)[Sign Up](#)

ACCOUNT TYPE

[Customer](#)[Admin](#)

USERNAME

PASSWORD

[Sign In](#)

GOURMET CONFECTIONS CHECKOUT

Place Your Order

SELECTED MASTERPIECE

Belgian Chocolate Cake

₹500

♦ CUSTOMER PARTICULARS ♦

Your Full Name *

Contact Number *

Delivery Date *



Complete Delivery Address *

[Confirm Order \(Cash on Delivery\)](#)[Cancel & Return to Showcase Menu](#)



Sweet Crumbs
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 admin
ADMINISTRATOR

Logout

[Store](#) [Dashboard](#)

Administrator

ADMINISTRATOR

Welcome, *admin*

Manage customer orders and the cake menu.



ACTIVE ORDERS
8
Awaiting fulfillment



DELIVERED
2
Successfully shipped



REVENUE
₹1798.00
From completed orders



ON MENU
5
Signature cakes

Customer Orders 8

Manage Cakes 5

**All Orders**
Change the status pill to update

SWEETCRUMBS.ORDERS

ORDER	CUSTOMER	MOBILE	ADDRESS	CAKE	QTY	PAYMENT	PLACED	STATUS
8	Bishal Saha	9635741490	Guma	Belgian Chocolate Cake	1	₹500.0	2026-07-18	BAKING
7	Bishal Saha	9635741490	Patuli	Chocolate Truffle Delight	1	₹899.0	2026-07-14	DELIVERED
6	Bishal	5632418964	Kolkata	Mango Delight Cake	1	₹499.0	2026-06-24	BAKING
5	Debansu	9163400137	123 Park Street, Near Metro Plaza, Kolkata, West Bengal 700016, India	Red Velvet Delight	1	₹899.0	2026-06-17	DELIVERED
4	Jacob	+915678942381	Patuli	Belgian Chocolate Cake	1	₹500.0	2026-06-09	RECEIVED
3	Jacob	+915678942381	Patuli	Belgian Chocolate Cake	1	₹500.0	2026-06-03	RECEIVED
2	Bishal Saha	9635741490	Habra	Belgian Chocolate Cake	1	₹500.0	2026-06-03	RECEIVED
1	Bishal Saha	9635741490	Habra	Selected Artisanal Cake	1	₹0.0	2026-06-03	RECEIVED

 Sweet Crumbs
ARTISAN BAKERY · PASTRY

Open Today
8 AM - 9 PM

Call Us
+91 98765 43210

 admin
ADMINISTRATOR

Logout

Remaining Items: 10

Successfully shipped

From completed orders

Signature: Cakes

Customer Orders 8

Manage Cakes 5

 ADMINISTRATOR · MENU

Add New Cake to Menu

Cake name will appear here...

IDENTITY

CAKE NAME *

e.g. Belgian Chocolate Cake

CATALOGUE N° *

e.g. 04

Displayed as N°04 on the cake card

FLAVOUR *

e.g. LEMON

Auto-uppercased on card corner

PRICING & RATING

BASE PRICE (₹) *

850

RATING (1-5) *

4.8

PRESENTATION

BADGE / TAG

None

Bestseller

New

Limited

Seasonal

CAKE IMAGE *

Choose File

No file chosen

Choose an image from your device

DESCRIPTION & INGREDIENTS

DESCRIPTION *

Write a short, enticing description of the cake...

KEY INGREDIENTS

Lemon zest, Butter cream, Edible flowers

Comma-separated — rendered as tags on the cake card

ADD CAKE TO MENU

PROJECT DIRECTORY

STRUCTURE

PAGE - 10

CakeShop/

|

|-- src/java/controller/ Servlets, DAO classes,

AuthUtil

|-- src/java/model/ Cake.java, User.java

|-- web/ JSP pages, CSS, assets,

WEB-INF

| |-- index.jsp Home page / cake catalogue

| |-- login.jsp Sign in and sign up

| |-- order.jsp Checkout (customer) / cake
details (admin)

| |-- my-orders.jsp Customer order history

| |-- my-account.jsp Customer profile

| |-- admin.jsp Admin dashboard

| |-- style.css Application stylesheet

| |-- assets/ Cake product images

| |-- WEB-INF/web.xml Deployment descriptor

|

1 CUSTOMER FEATURES

- Browse cake catalogue with images, ratings, descriptions, ingredients
- Register account (username, email, password, full name, phone)
- Sign in and sign out
- Buy a cake (redirects to login if not authenticated)
- Complete checkout with name, phone, delivery date, and address
- View order history at /my-orders
- View account details at /my-account
- Track order status updates from admin

2 ADMINISTRATOR FEATURES

- Sign in with admin credentials
- Access Admin Console at /view-orders
- View dashboard stats: active orders, delivered count, revenue, menu count
- Review all customer orders with delivery details
- Update order status via dropdown
- Add new cakes to the menu
- Click Buy on home page to view full cake details (read-only)

1 CUSTOMER ORDER WORKFLOW

Step 1 : Customer visits home page and browses cakes

Step 2 : Customer clicks Buy on a desired cake

Step 3 : If not logged in -> redirected to login with return URL

Step 4 : After login -> checkout page with pre-filled name and phone

Step 5 : Customer submits order -> saved to database with user_id

Step 6 : Customer redirected to My Orders with success message

Step 7 : Admin updates order status as it progresses

Step 8 : Customer sees updated status in My Orders

2 ADMIN WORKFLOW

Step 1 : Admin goes to login and selects Admin account type

Step 2 : Admin enters credentials and opens Admin Console

Step 3 : Admin reviews incoming orders on Customer Orders tab

Step 4 : Admin updates status (Received -> Baking -> Out for Delivery -> Delivered)

Step 5 : Admin switches to Manage Cakes tab to add new products

Step 6 : Admin can browse home page and click Buy to inspect cake details

ORDER STATUS LIFECYCLE:

Valid statuses (managed by OrderDao):

1. Received (default for new orders)
2. Baking
3. Out for Delivery
4. Delivered
5. Cancelled

Admin updates status from the Admin Console.

Customers see current status on the My Orders page.

PREREQUISITES:

- Java JDK 8 or higher
- Apache NetBeans IDE with Java EE support
- GlassFish Server (Java EE 7)
- MySQL Server on localhost:3306
- MySQL Connector/J JAR (mysql-connector-java-5.1.49.jar)

- Password hashing (BCrypt) for secure storage
- Payment gateway integration (UPI, cards)
- Email/SMS order confirmation notifications
- Cake image upload from admin panel
- Edit and delete existing cakes
- Customer order cancellation before baking
- Search and filter cakes by category or price
- Connection pooling (e.g. HikariCP)

Conclusion:

CakeShop is a complete bakery e-commerce web application built with Java, Servlets and JSP. It demonstrates session-based authentication, JDBC database operations, role-based access control, and a full order management workflow.

The system is suitable for academic projects or small business use with further security improvements such as password hashing.

Teacher's Signature